

ME350 FA 2025 Section 002

Lab #5

Monday 12:30 PM - 2:30 PM

Haoran Shi

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)



Reminders and Project Schedule

- *Tue* Sep. 30 MIDTERM EXAM
- *Mon* Oct. 06 Gate Review 2 in Lab 6
- *Week of* Oct. 13 Functional Testing outside of lab
(linkage parts + hardstops due)



Look Ahead

Lab 6: Gate Review 2 (X50 Assembly Room)

- Review complete BOM, transmission selection, transmission CAD, limit switch mounting in CAD

Fall break! Lab 7: Functional Testing (X50 Assembly Room)

- Try your mechanism (four links, spacers, hard stops, baseplate) on the arena during free time
- Mechanism should move through its range of motion and hit all 4 photosensors without collisions

Kits Are Ready!

- Project kits are located in the cabinets underneath workbenches and the vertical cabinets along the wall in the X50 Assembly Room and are labeled by team number
- Electronic components will be provided to your team later in the semester
- Kits should include:
 - Sleeve Bushings
 - Needle Thrust Bearings
 - Plate, Angle, and Round Aluminum Stock
 - Motor and Motor Bracket
 - Flashlight
- If you don't have M-Card access to the X50 Assembly Room, contact Don Wirkner (dwirkner@umich.edu)



Calendar - Key Dates

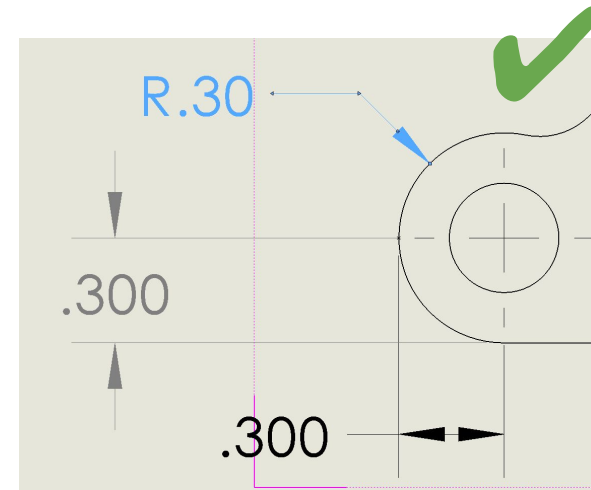
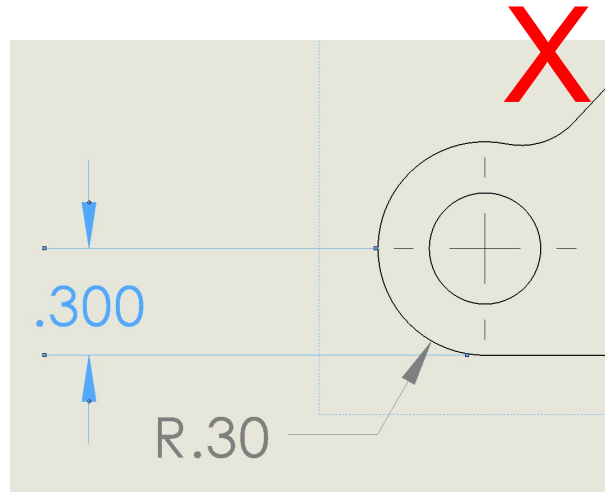
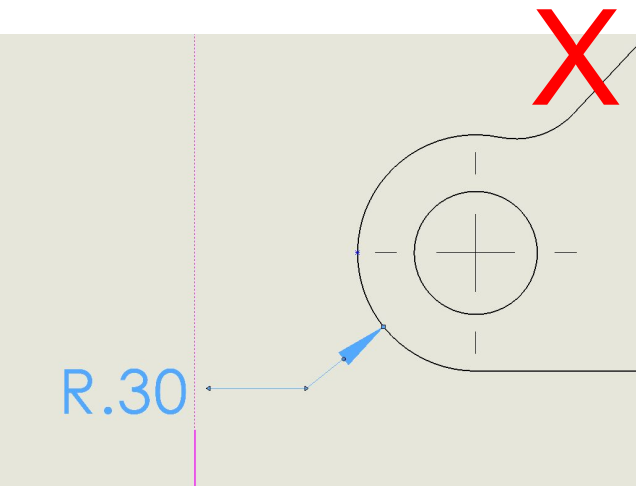
Sep 28	Sep 29	Sep 30	Oct 1	Oct 2	Oct 3
		Midterm Exam		Lecture 11: Selection of Motor and Transmission; Conflict Resolution (MU)	
					Homework 4 Released
	Lab 5 - Transmission Selection Tutorial			Lab 6 - Gate Review 2 in Lab (X50 Assembly Room, 1510 GGB) Lab Assignment #5 Due - Transmission Selection Workbook Bill of Materials (BOM) Transmission CAD	Lab 6 - Gate Review 2 in Lab (X50 Assembly Room, 1510 GGB) Lab Assignment #5 Due - Transmission Selection Workbook Bill of Materials (BOM) Transmission CAD
				Build & Assembly for Functional Testing -->	
Oct 5	Oct 6	Oct 7	Oct 8	Oct 9	Oct 10
		Lecture 12 - Transmission Gear Forces (MU)		Lecture 13 - Transmissions: Gears - Strength and Wear (MU)	
	Lab 6 - Gate Review 2 in Lab (X50 Assembly Room, 1510 GGB) Lab Assignment #5 Due - Transmission Selection Workbook Bill of Materials (BOM) Transmission CAD			Lab 7 - Functional Testing (X50 Assembly Room, 1510 GGB) Lab Assignment #6 Due - Manufactured GR1 Parts	Lab 7 - Functional Testing (X50 Assembly Room, 1510 GGB) Lab Assignment #6 Due - Manufactured GR1 Parts
				<-- Build & Assembly for Functional Testing	
Oct 12	Oct 13	Oct 14	Oct 15	Oct 16	Oct 17
		No Lecture -- Fall Study Break		Lecture 14 - Intro to Mechatronics (JK) - Homework 4 Due (Transmissions 1) - Homework 5 Released	
	No Lab -- Fall Study Break Lab 7 - Functional Testing TO BE COMPLETED ASYNCHRONOUSLY FOR MONDAY LABS Lab Assignment #6 Due - Manufactured GR1 Parts			Lab 8 - Sensors and Microcontrollers 1: Wiring, switches, and IR Prox sensor Lab Assignment #7 Due - Manufactured GR2 Parts	Lab 8 - Sensors and Microcontrollers 1: Wiring, switches, and IR Prox sensor Lab Assignment #7 Due - Manufactured GR2 Parts

General Notes for Drawings/MPs

- **Dimension from one hole to the others**
 - After you get to the first hole, consider rezeroing the mill
 - It is helpful for your drawing to dimension from the first hole
- **Waterjet clearance holes**
 - If you have a clearance hole for a loose fit bolt, look up the correct clearance hole size for the bolt and then make the waterjet do the work
- **Waterjet drawings**
 - If the part is very complicated, use 2 sheets for one drawing with different shown dimensions. Try to include overall part dimensions on both and **add a note to the Comments Block.**
- **You need a printed drawing/manufacturing plan with Don or Kemal's signature to work in the shop**
 - Don't print out a non-signed version
 - Don't try to pull it up on your phone
 - Laptops or tablets OK (need a big enough screen)

General Notes for Drawings/MPs

- Fully dimension your holes at the center of a radius



Manufacturing Demo Videos

- https://drive.google.com/drive/folders/1I_ROVWrFD5i1GvIyN3_0y9uKB_ddeNIs
- Available on student intranet as well



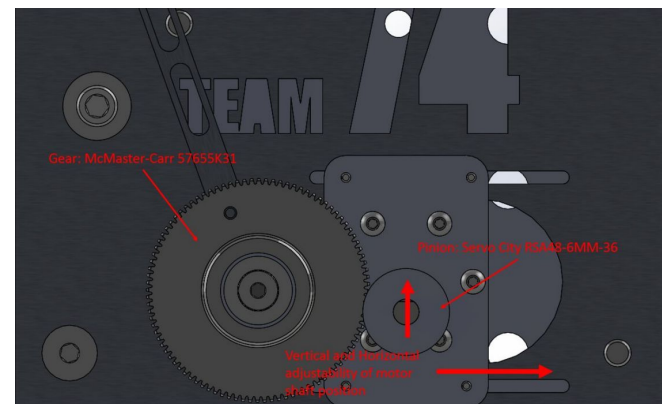
3D Printing Guidelines

- **3D Printing in the Duderstadt:**
 - 3D Printers (PLA FDM only) are available in the Duderstadt Fabrication Center on the ground of the Duderstadt, behind the library desk
 - Reference the [3D Printing Guide](#) to get certified and schedule a 3D Printing Slot
 - Complete the [Certification on this Canvas Page](#)
 - You can reserve time on the Prusas using the [Student Signup Sheet](#).
 - 3 hour time slots five times a day. You can sign up for a slot at any time the printers are available.
 - ME 350 has registered a certain number of hours. Notify caffaki@umich.edu if there are no more available hours for ME 350.
 - **Access code: L5KI32**
- **3D Printing in the Assembly Room**
 - 3D Printers are available in the 250 Assembly Room
 - Your sliced print file must be approved by Kemal Duran (keduran@umich.edu)
 - You can reserve time on the Bambus using _____

Transmission Design

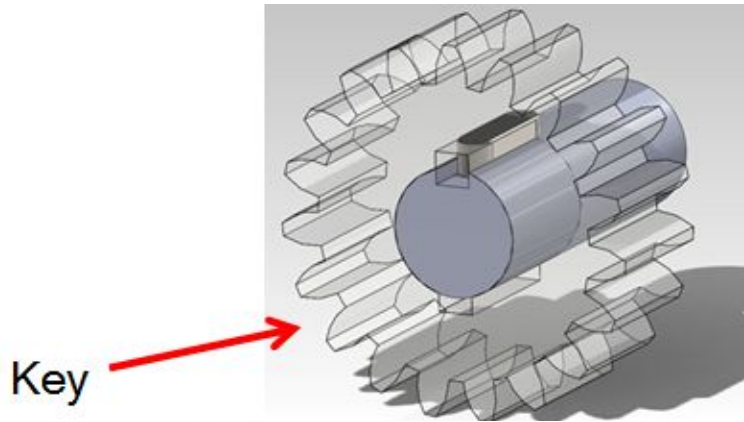
- Model the entire transmission system in CAD
- To be complete, the final CAD model should include:
 - Motor & mounting
 - Torque transferring joints - show in the CAD model how torque is transferred
- Take screenshots
 - Make sure all relevant features are properly labeled and clear to audience (i.e. someone who has not built a mechanism but is familiar with the project) – factors into your grade for “Quality of Report”
- Estimate weight of the moving components of your mechanism

Example screenshot:

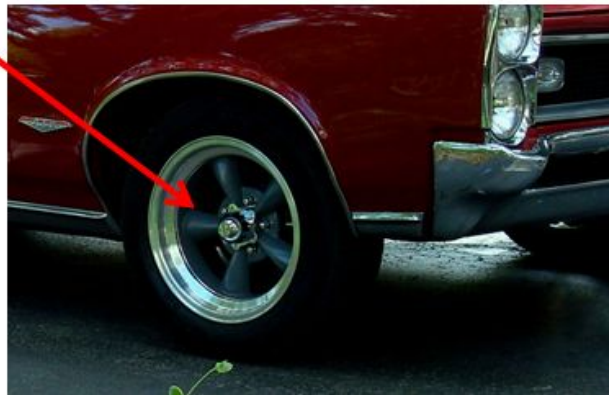


Torque Transfer

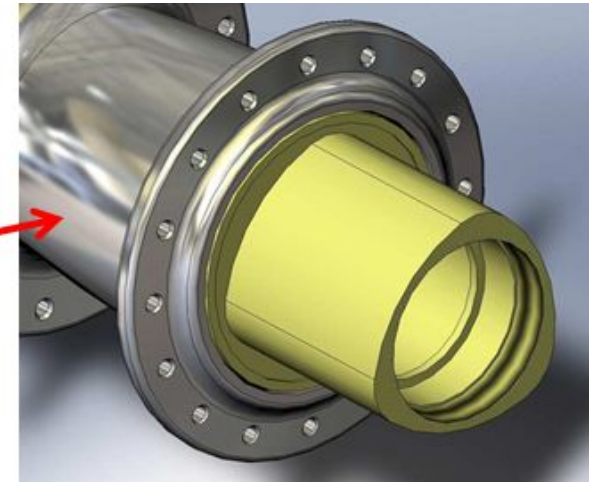
How do you effectively transfer rotary power?



Wheel studs are an example of multiple off-axis pins



Polygon



Torque Transfer

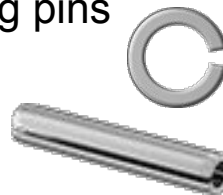
How do you effectively transfer rotary power?

- Set screws are good in low torque, but bad for high torque applications
- Off-axis pins are easy to integrate for low torque
- Splines/Polygons are good in higher torque applications
 - Often expensive/difficult to make, industry standard
- Keys are for moderate to high torque applications
 - Easy to make, load capability proportionate to key size

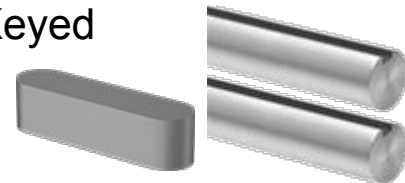
Set Screws



Spring pins



Key + Keyed shaft



Spline



Torque Transfer

- Develop methods of torque transfer (pin, set screw, etc...)
- Identify joints in which torque is transmitted
- Show how torque is transferred
- Show screenshots of cross section of the joints
- Perform analysis at torque transmitting joints
 - HINT: Perform analysis to determine if a set screw (i.e. set screw sitting on a motor shaft) will slip. Use the following reference material:
 - [Link to coefficients of friction](#), [Table](#), [deep dive into fastener theory](#)
 - HINT: Perform analysis to determine if a dowel pin or spring pin used to connect your gear to your link will shear
- Only on splines, off-axis pins, keys, or set screws (not needed on gear hubs, shafts, links, etc.)
- NO FEA! (use methods taught in 211, 250, 382, 350)

Transmission Ratio

Find transmission ratios using two different methods:

- Inertia matching: to maximize acceleration torque
- Resolution at the flashlight: desired positioning resolution of flashlight

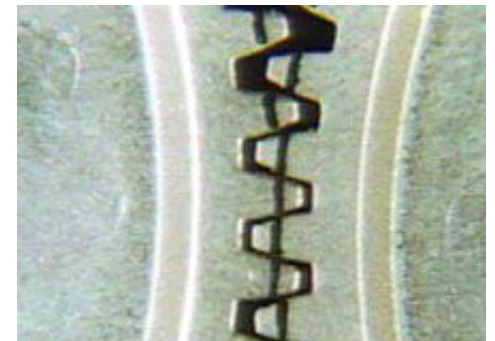
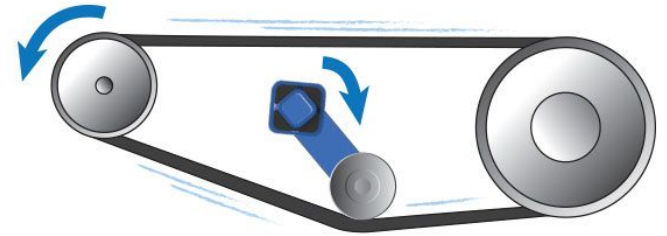
Choose the most conservative one to be your final transmission ratio

- **The largest one! (Largest speed reduction)**

Transmission Design

Motor mounts

- Belts/chains stretch over time and need to be held in tension
- Gear mesh requires accurate machining and tight tolerances
- Center-to-center holes are a critical feature!



Transmission interface

- Motor side: Check out 2D & 3D CAD of pulleys/gears
 - Compare with motor shaft dimension (**0.25" vs 6 mm**)
 - Flange vs. flangeless? (for belts)
 - Loading on motor/linkage bearings
- Linkage side: Customize interface
 - Need to transmit torque!

Transmission Design

Use a systematic way to select a final type of transmission

- Belts, gears, chains, lead screws
- One option is to use a Pugh chart or similar method with a weighting scheme
- Clearly define criteria & associated weights
- Provide full justification of your criteria & weighting scheme

Select parts from McMaster, SD-PSI, Misumi, ServoCity, or other websites and note

- (1) manufacturer/website
- (2) part #
- (3) unit cost for each transmission component



Limit Switch Mounting

- Consider how you are going to attach the limit switch(s) to your linkage/baseplate.
 - It must contact a link (input or follower) at one end of the full range of motion. You can use two limit switches (one for each limit) to improve encoder accuracy.
 - The switch need to compress before your linkage hits a hardstop (consider adjustability)
 - The [CAD is available here](#)



Today

- Determine your transmission ratio by following the [Lab 5 Tutorial](#)
- Work on Drawings and Manufacturing Plans
- Start filling out your complete Bill of Materials
- Ask me design or manufacturing questions

Due next lab:

1. [Transmission Selection Workbook](#)
2. Transmission CAD
3. [Complete BOM](#)

**Next two labs will be in the X50 Assembly Room (1510 GGB),
bring safety glasses!**

